

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Capstone Project	Aim: System Design and Architecture	
	Date: 25/09/25	Enrolment No: 92200133037

Introduction

InterviewBot is a web-based application designed to simulate realistic job interviews, leveraging AI to generate tailored questions, provide real-time feedback, and produce comprehensive performance summaries. The system integrates a Flask backend, HTML/JavaScript frontend, and external APIs (Google Gemini for question/feedback generation, ElevenLabs for text-to-speech) to deliver an immersive experience with a 3D animated interviewer avatar, resume processing, session recording, and PDF report generation. The design emphasizes modularity, a robust technology stack, and scalability to meet stakeholder needs, including candidates seeking effective interview practice and recruiters requiring reliable evaluations.

This section outlines the system's modular design, technology stack, and scalability plan, supported by a system architecture diagram. The design ensures maintainability, performance, and extensibility, aligning with the project's objectives to provide a user-friendly, scalable interview simulation platform. The system is currently tested locally and prepared for deployment on Render, with scalability considerations for cloud hosting.

Modular Design

Modular Structure

InterviewBot is architected with a modular approach, dividing the system into distinct, independent components to enhance maintainability, reusability, and extensibility. The system comprises four primary modules:

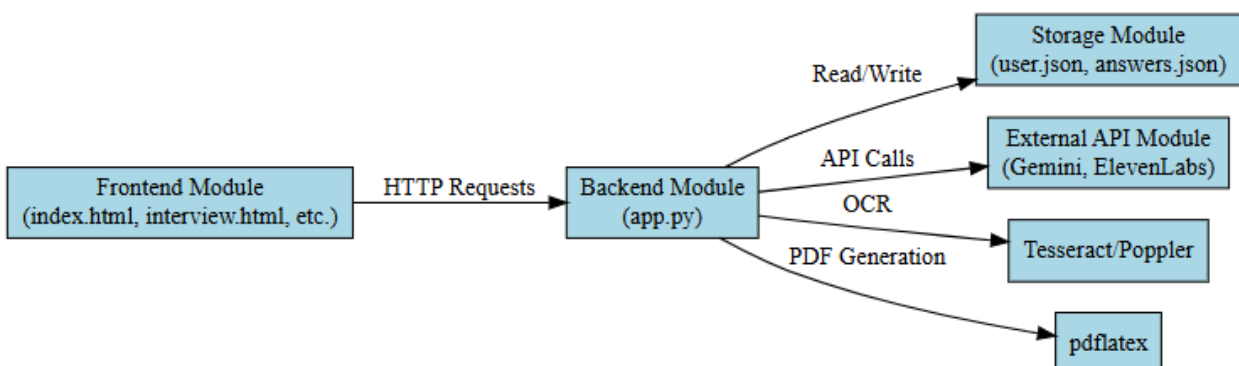
1. Frontend Module:
 - **Functionality:** Handles user interactions via HTML pages (index.html, rules.html, interview.html, summary.html, screen.html). Provides form submission, 3D avatar rendering, webcam integration, speech recognition, and report display.
 - **Subcomponents:**
 - **User Interface (UI):** HTML/CSS for responsive layouts.
 - **Client-Side Logic:** JavaScript (scripts.js) for API calls, Three.js for 3D avatar, Web Speech API for speech recognition, and MediaRecorder for video capture.
 - **Purpose:** Ensures a seamless user experience with minimal dependencies on backend logic.
2. Backend Module:

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- **Functionality:** Manages core logic, including API endpoints (app.py) for data processing, question generation, feedback, TTS, and PDF creation.
 - **Subcomponents:**
 - **API Layer:** Flask endpoints (/submit_user_data, /generate_questions, /tts, /generate_answer_feedback, /store_answers, /generate_summary, /generate_pdf).
 - **Processing Layer:** Handles resume OCR (Tesseract/Poppler), JSON storage (user.json, answers.json), and LaTeX compilation (pdflatex).
 - **Purpose:** Centralizes business logic, ensuring modularity through endpoint separation.
3. **Storage Module:**
- **Functionality:** Uses JSON files (user.json, answers.json, interview_report.json) for temporary data persistence, storing user details, answers, and reports.
 - **Purpose:** Provides lightweight, file-based storage, easily replaceable with a database for scalability.
4. **External API Module:**
- **Functionality:** Integrates Google Gemini for question/feedback generation and ElevenLabs for TTS, with gTTS as a fallback.
 - **Subcomponents:**
 - **Gemini Integration:** Generates tailored questions and feedback.
 - **ElevenLabs/gTTS Integration:** Converts text to audio for question playback.
 - **Purpose:** Offloads complex computations to external services, reducing local processing overhead.

System Architecture Diagram

The following diagram illustrates the modular structure and interactions:



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Modularity Benefits

- **Maintainability:** Each module can be updated independently (e.g., modifying interview.html without affecting app.py).
- **Reusability:** Components like TTS or OCR logic can be reused in other projects.
- **Extensibility:** Adding new features (e.g., database storage, multi-language support) requires minimal changes due to modular endpoints and JSON storage.

Technology Stack

Technology Selection

The technology stack was chosen based on project requirements for performance, compatibility, and ease of development:

1. Programming Languages:

- **Python 3.10:** Used for the backend due to its simplicity, extensive libraries, and compatibility with AI/ML tools (e.g., Tesseract, Gemini). Justified by Python's dominance in web and AI applications (Python Software Foundation, 2023).
- **JavaScript/HTML5/CSS3:** Used for the frontend for cross-browser compatibility and dynamic interactions. JavaScript's Web Speech API and Three.js enable real-time speech recognition and 3D rendering.

2. Frameworks and Libraries:

- **Flask 2.3.3:** Lightweight web framework for Python, ideal for rapid API development. Chosen over Django for its simplicity and lower overhead for small-scale applications (Flask Documentation, 2023).
- **Gunicorn 22.0.0:** WSGI server for deploying Flask, supporting concurrent requests (Gunicorn Docs, 2023).
- **Three.js:** Enables 3D avatar rendering in interview.html, chosen for its robust WebGL support (Three.js Documentation, 2023).
- **Pillow 10.4.0:** Processes images for OCR, selected for its reliability in Python-based image manipulation.
- **pytesseract 0.3.13:** Interfaces with Tesseract for resume text extraction, chosen for open-source availability and accuracy (Tesseract OCR, 2023).
- **pdf2image 1.17.0:** Converts PDFs to images for OCR, compatible with Poppler (Poppler Docs, 2023).

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- pandas 2.2.2: Analyzes OCR output for name extraction, selected for efficient data handling.
- requests 2.32.3: Handles API calls to Gemini and ElevenLabs, chosen for its simplicity.
- gTTS 2.5.3: Fallback TTS, ensuring offline functionality.
- flask-cors 4.0.1: Manages cross-origin requests for local testing.

3. External Services:

- Google Gemini API: Generates questions and feedback, chosen for its advanced NLP capabilities and ease of integration (Google Cloud, 2025).
- ElevenLabs API: Provides high-quality TTS, selected for realistic voice output (ElevenLabs Docs, 2025).

4. System Dependencies:

- Tesseract 5.3.0: Open-source OCR engine for resume processing.
 - Poppler 22.12: PDF rendering for OCR.
 - TeX Live 2023: Compiles LaTeX for PDF reports, chosen for professional document formatting.
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- **Performance:** Flask and Gunicorn ensure low-latency API responses (<3s). Gemini and ElevenLabs offload heavy computations, reducing local resource usage.
 - **Compatibility:** Python libraries integrate seamlessly with Flask, while JavaScript supports modern browsers.
 - **Stakeholder Needs:** Candidates require an intuitive UI (HTML/JS) and realistic interviews (Gemini/ElevenLabs). Recruiters benefit from accurate reports (pdflatex).
 - **Cost:** Open-source tools (Flask, Tesseract) minimize costs, while Gemini/ElevenLabs APIs are usage-based, suitable for Render deployment.

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InterviewBot's system design leverages a modular architecture, a robust technology stack, and a comprehensive scalability plan to deliver a reliable, extensible interview simulation platform. The modular design enhances maintainability, the technology stack ensures performance and compatibility, and the scalability plan prepares the system for cloud deployment on Render.